

1 · OFFLINE ARTIFACTS

isical

calibration studio
ChArUco + AprilGrid
→ bundle adjustment



calibration.json

isiGen + isidet

synthetic data
(SDXL + ControlNet + LoRA)
→ detector training



model.onnx

2 · RUNTIME NODE (one PC, one GPU)

cameras

2× RTSP
H.264 / H.265

frames
/dev/shm bus

RTSP

isistream

perception producer
capture · NVDEC
zone-scoped detection
pose

UDP :9010

monitor_web

operator dashboard
START / STOP
supervision

tracks · zones
UDP :9001

backbone

metric engine
homography · triangulation
ByteTrack-in-meters
gating

detection sets

3 · DELIVERY

consumers

AGVs · dashboards

REST poll

isicomms gateway

aggregation
REST :8080 · probe UI

subscribe

MQTT broker

Mosquitto :1883

MQTT
schema v6

solid blue = live data plane (wire contracts) · dashed = file artifacts / shared memory · gray = inputs